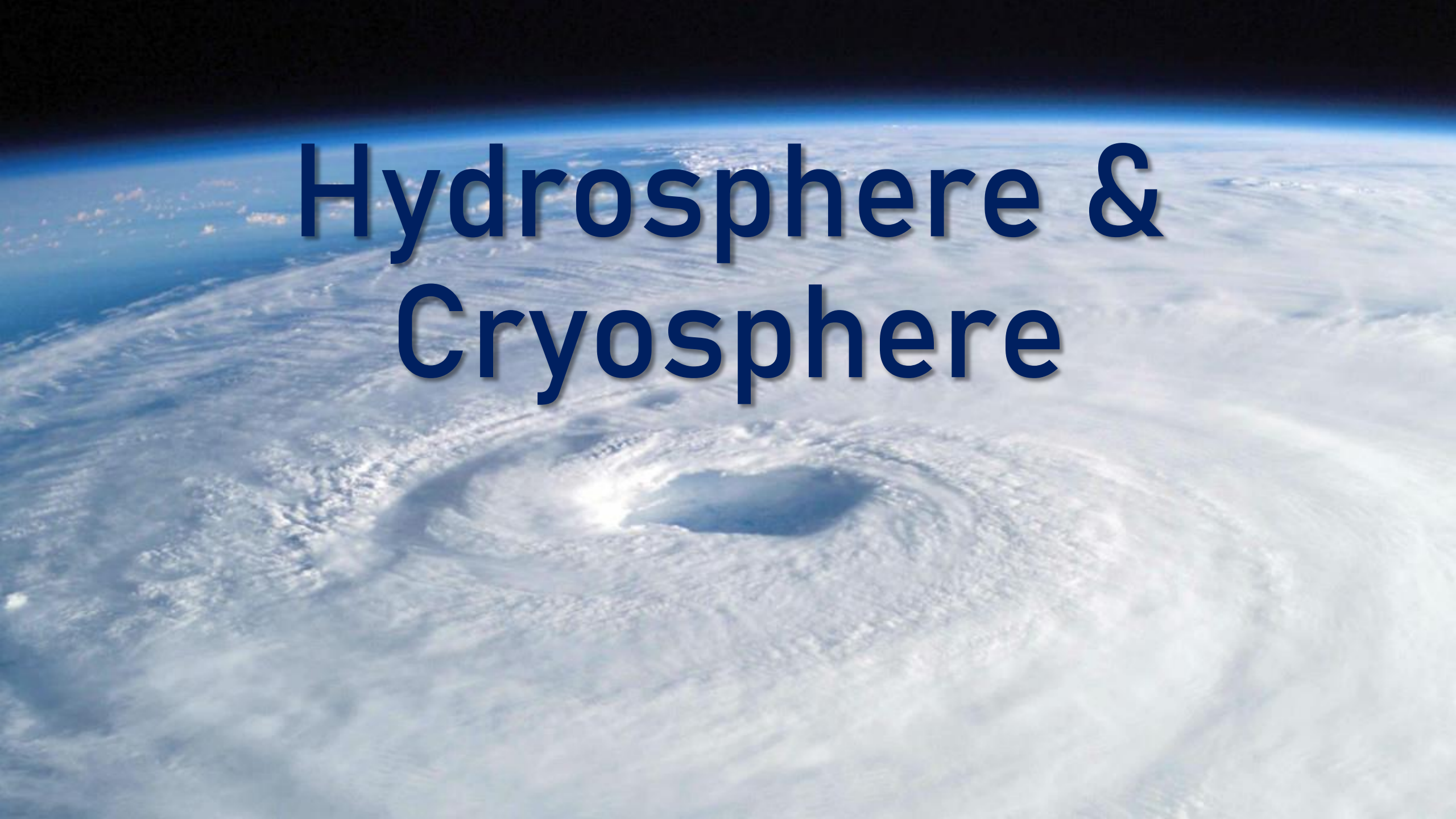


Hydrosphere & Cryosphere

A satellite image of Earth showing a large portion of the Arctic region. The central feature is a dark blue, circular body of water, likely the Beaufort Sea, surrounded by a vast expanse of white and light blue sea ice. The ice shows intricate patterns of ridges, leads, and floes. The horizon of the Earth is visible at the top, with a thin blue line of the atmosphere against the black background of space.

Summary of previous class

Earth's Layers and Structure

- What are the four main layers of the Earth from the outside to the inside?

Crust → Mantle → Outer Core → Inner Core

- Which layer of the Earth is the thinnest?

The **crust** is the thinnest of Earth's four main layers. (Ocean crust is thinner than continental crust)

- How do temperature and pressure change as you go deeper toward the center of the Earth?

Both **temperature and pressure increase** as you travel deeper into Earth.

- What is the difference between the solid inner core and liquid outer core?

The **inner core** is solid because of the **extremely high pressure**, while the **outer core** is **liquid** and made mostly of molten iron and nickel. The **outer core's moving liquid metal** produces Earth's magnetic field.



Summary of previous class

Plate Tectonics and Movement

- What is the lithosphere, and how does it relate to tectonic plates?

The **lithosphere** is Earth's rigid outer shell. The lithosphere is broken into large pieces called tectonic plates. These plates move slowly over a softer, hotter layer beneath them called the asthenosphere.

- What causes tectonic plates to move around?

Tectonic plates move because of heat and movement inside Earth. The three main forces are: mantle convection, ridge push, and slab pull. Slab pull is considered one of the strongest forces driving plate movement.

- What happens at a divergent plate boundary?

At a **divergent boundary**, two tectonic plates **move away from each other**. Divergent boundaries commonly occur at **mid-ocean ridges** under the oceans. Magma rises from beneath the plates, cools and forms new crust. On continents, divergent movement can create **rift valleys**.

- How do mountains form when two continental plates collide?

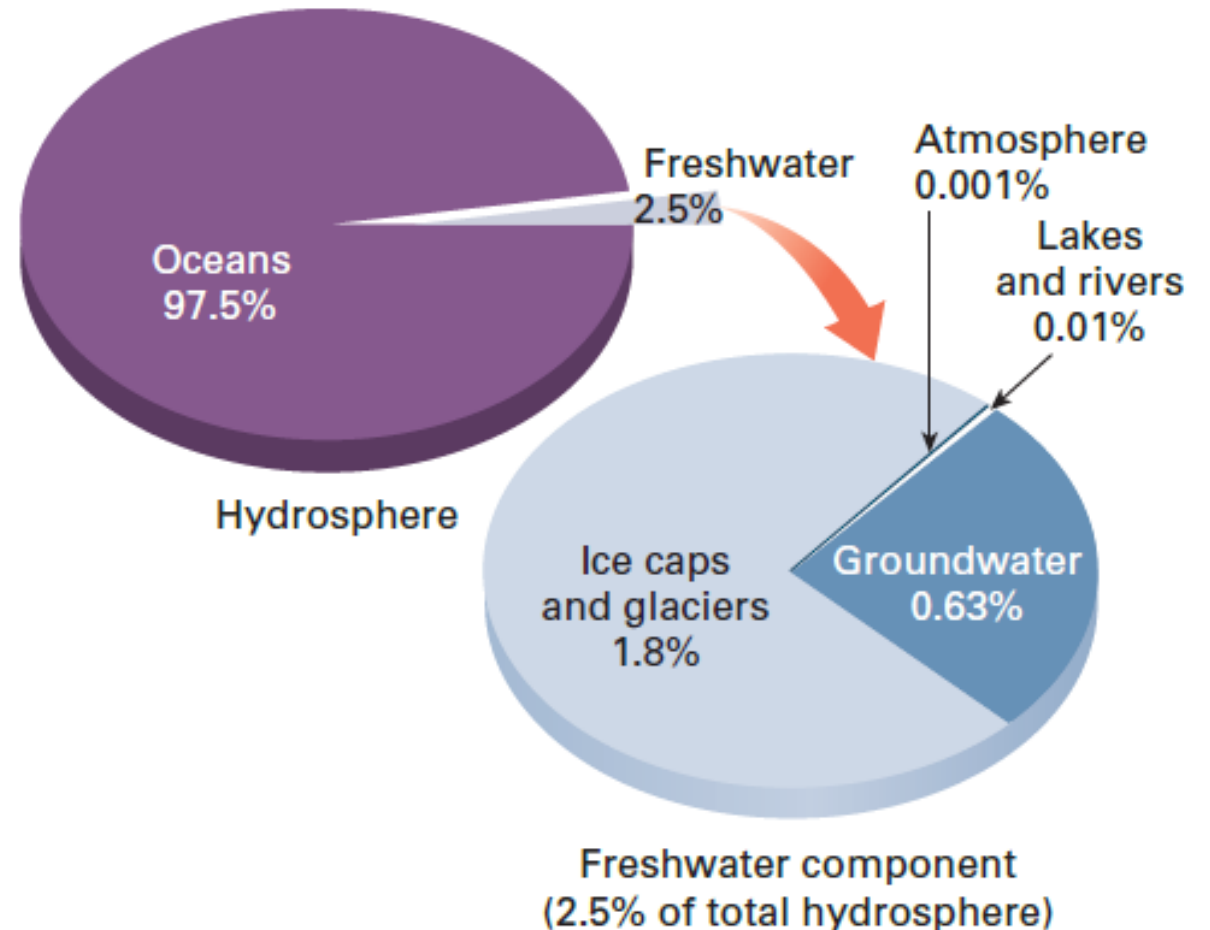
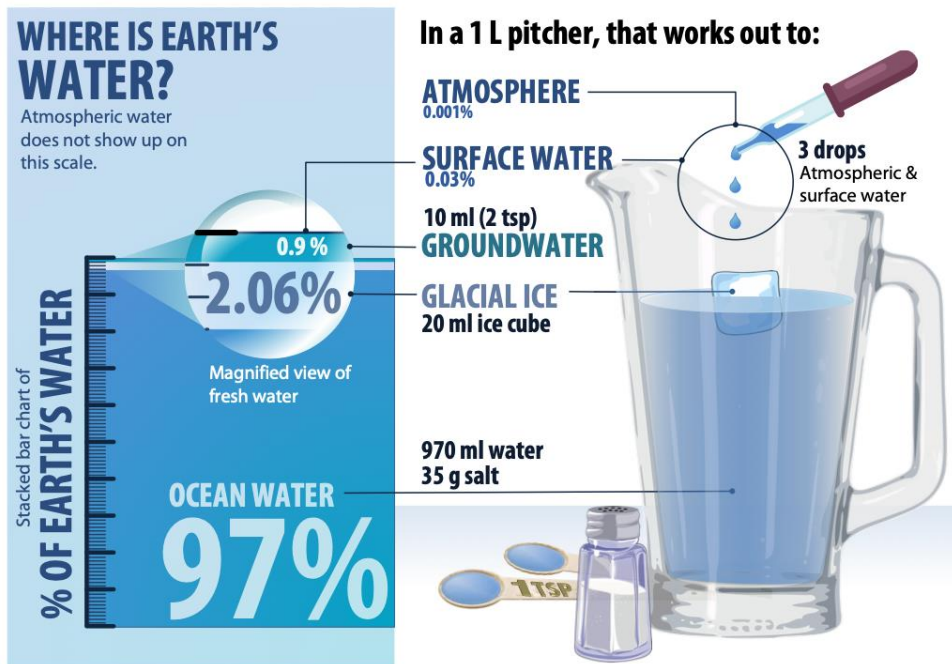
When **two continental plates collide**, neither plate easily sinks because continental crust is relatively light and buoyant.

- Instead:
 - The plates push against each other.
 - The crust becomes compressed, folded, and faulted.
 - The crust thickens and is pushed upward.
 - Large **mountain ranges** form.
- **Example:** The **Himalayas** formed because the Indian Plate collided with the Eurasian Plate.



Hydrosphere

- The hydrosphere encompasses **all the water present on Earth**, whether in liquid, solid, or gaseous form.
- It plays a vital role in shaping Earth's surface, influencing climate and weather patterns, and supporting life.



Types of water

1. SALTWATER



Found mainly in oceans and seas; it makes up about **97%** of Earth's water.



2. FRESHWATER



Makes up about **3%** of Earth's water and is found in glaciers, ice caps, groundwater, lakes, and rivers.



3. SURFACE WATER



Water found on Earth's surface, such as rivers, lakes, and ponds.



4. GROUNDWATER



Water stored beneath Earth's surface in soil and rocks.

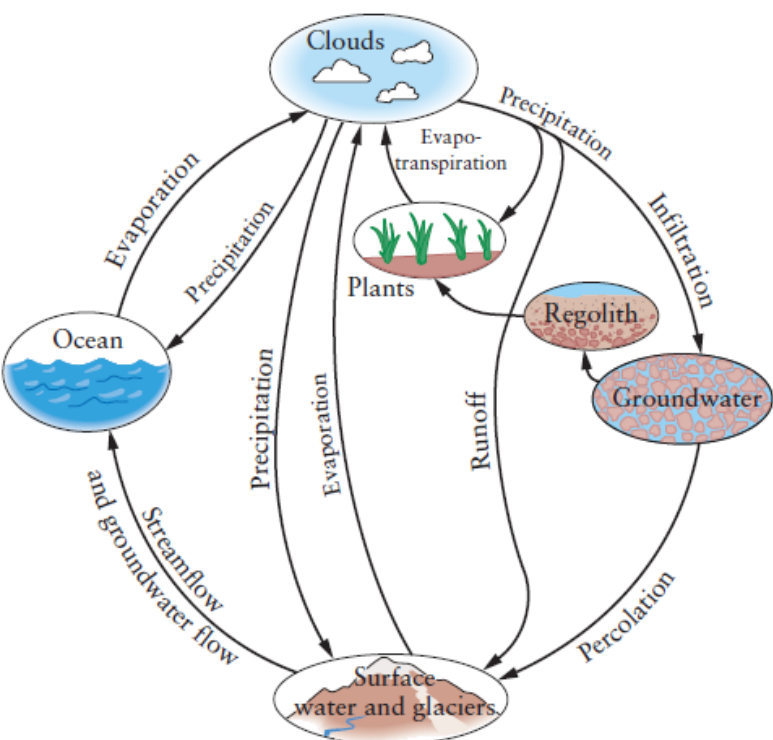


5. WATER VAPOR

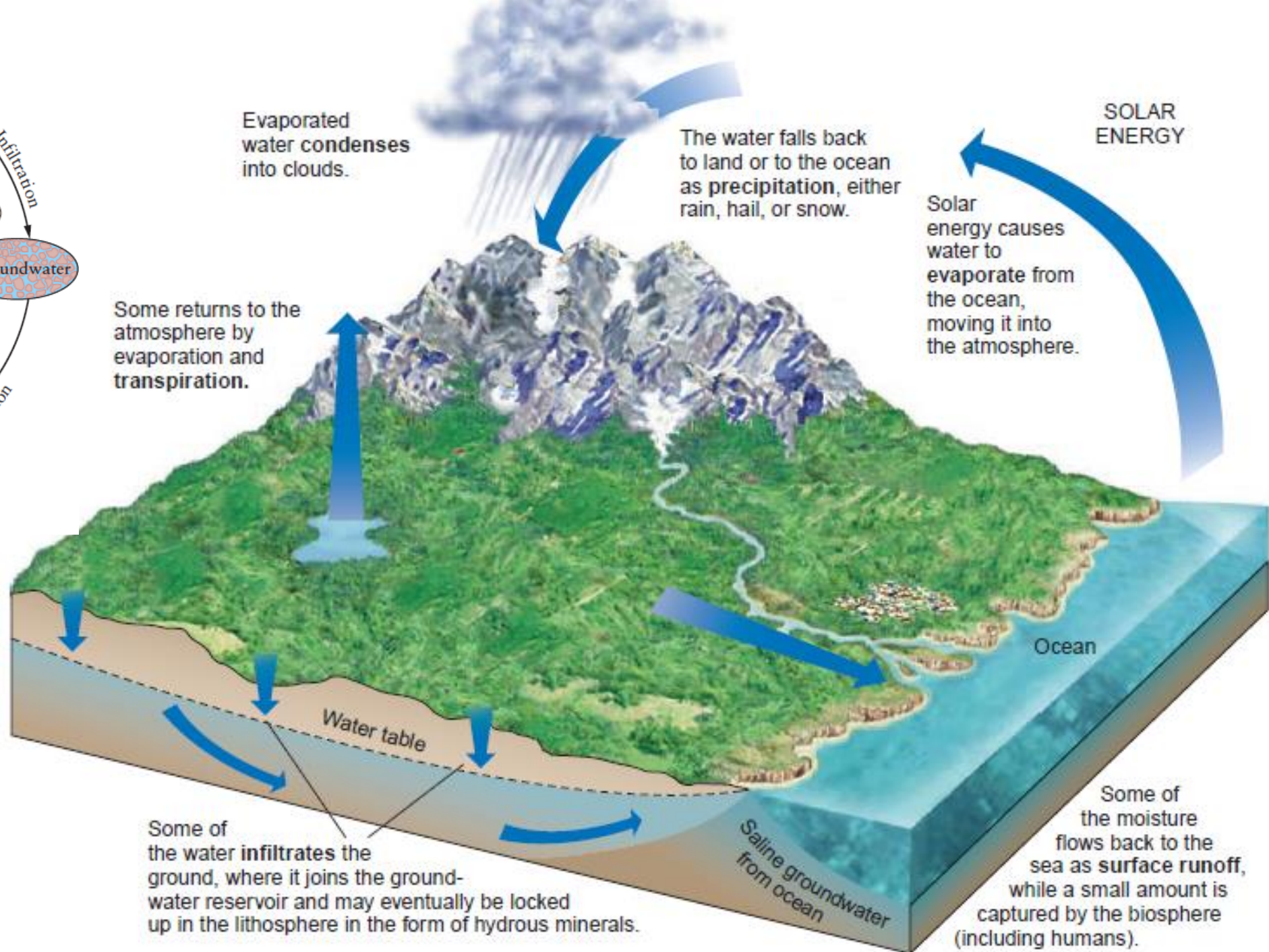


Water in gas form found in the atmosphere.





HYDROLOGIC CYCLE



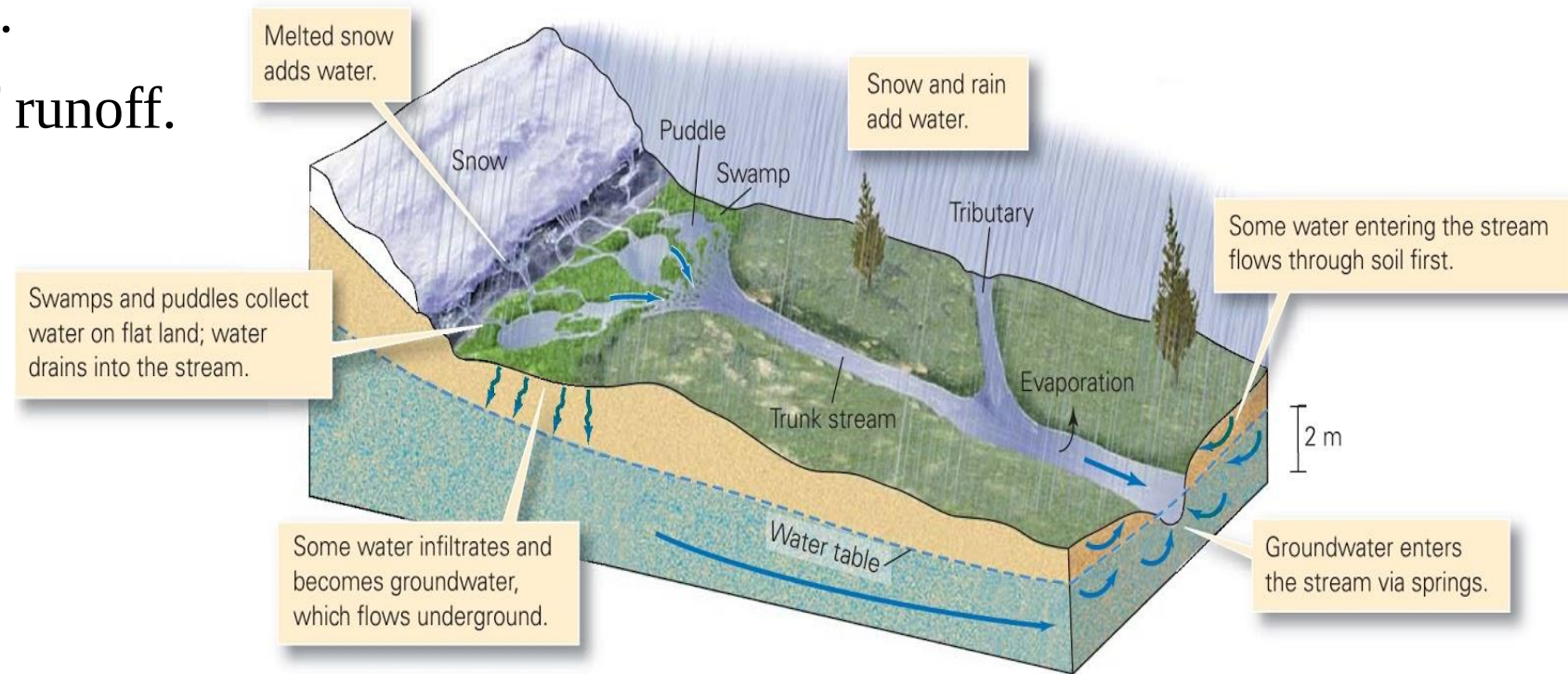
Origin of Streams (Running water)

- Stream flow begins as scattered *runoff*. The term **stream** is used for all water flowing in a channel, regardless of the stream's size from **high elevation to low elevation**.

How long does it take for water that falls as precipitation to reach the ocean?

- On average, **the residence time, the length of time that a given volume of water remains in streams**, is about 14 days.
- Rate of infiltration < Rate of runoff.

**Vol. water in
Groundwater ~ = 70x Vol.
water in Streams**



As this water flows it does the work of both erosion and deposition.

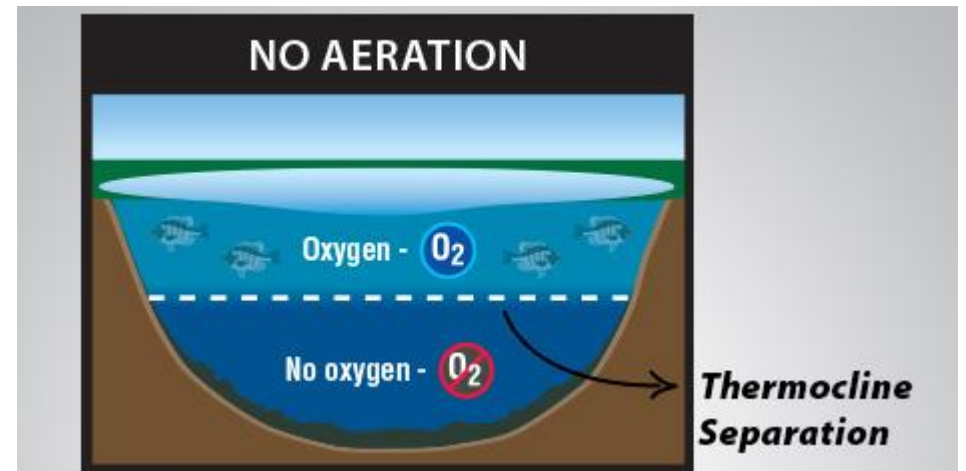


Lakes & Ponds

- A lake is a body of **inland water** of appreciable size with an **open surface**, occupying a depression in Earth's surface.
- A **smaller** body of fresh standing water is called a **pond**. A majority of the world's lakes are found in high latitudes and in mountains.
- Lakes are predominantly **fed by runoff** and direct **precipitation**, and their level and salinity commonly reflect a balance between freshwater input from streams and direct precipitation, outflow, and evaporation.
- In deep lakes – improper distribution of sunlight which leads to low level of biological productivity often causes to have a deep-blue color that we associate with a clean, healthy lake. Such a lake is called **oligotrophic**, meaning “**poorly nourished**.”

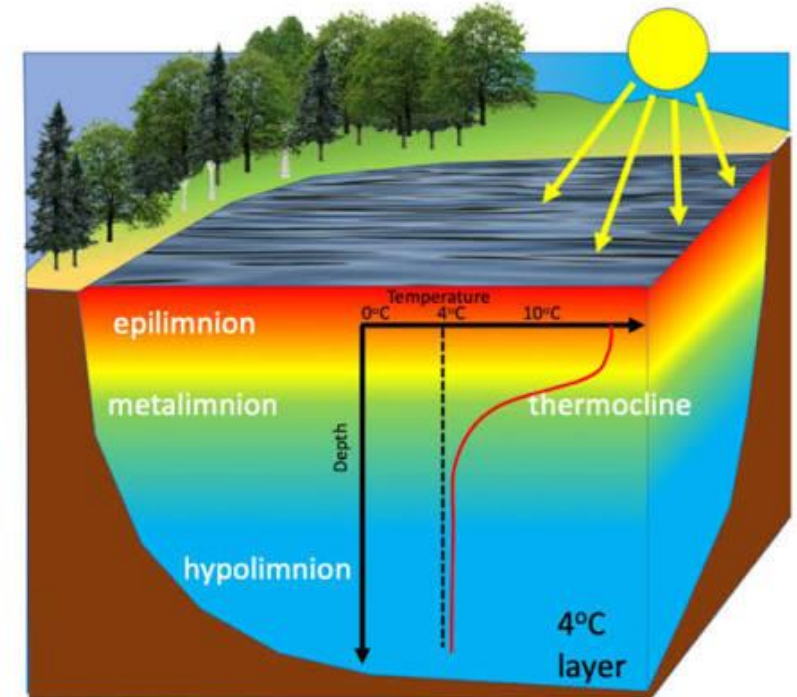
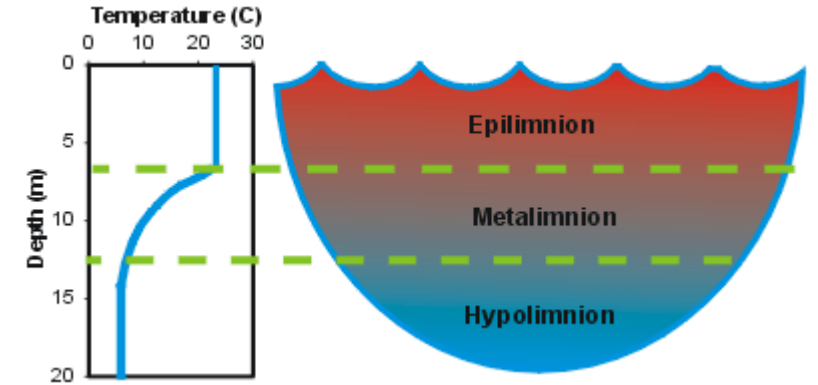


Wetlands are surface water bodies that are intermittently or seasonally wet.

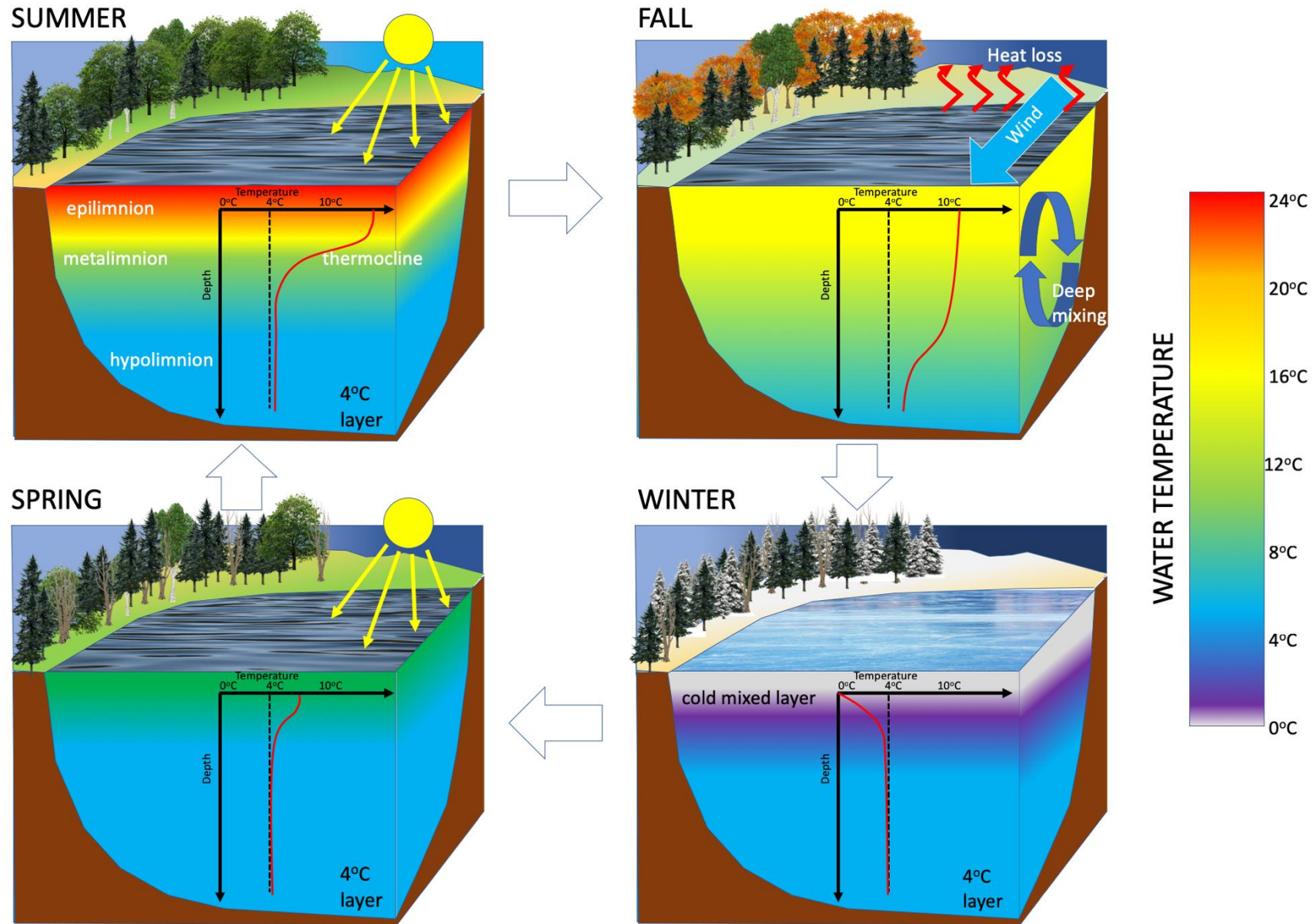


Thermal stratification of lakes

- **Thermal stratification** is the layering of lake water into different temperature zones because warm water is less dense than cold water.
- The three main layers are:
 - **Epilimnion** – The **warm upper layer**, heated by sunlight and usually well mixed.
 - **Metalimnion (Thermocline)** – The **middle layer**, where temperature changes rapidly with depth.
 - **Hypolimnion** – The **cold, deep layer**, which is usually denser and less mixed.

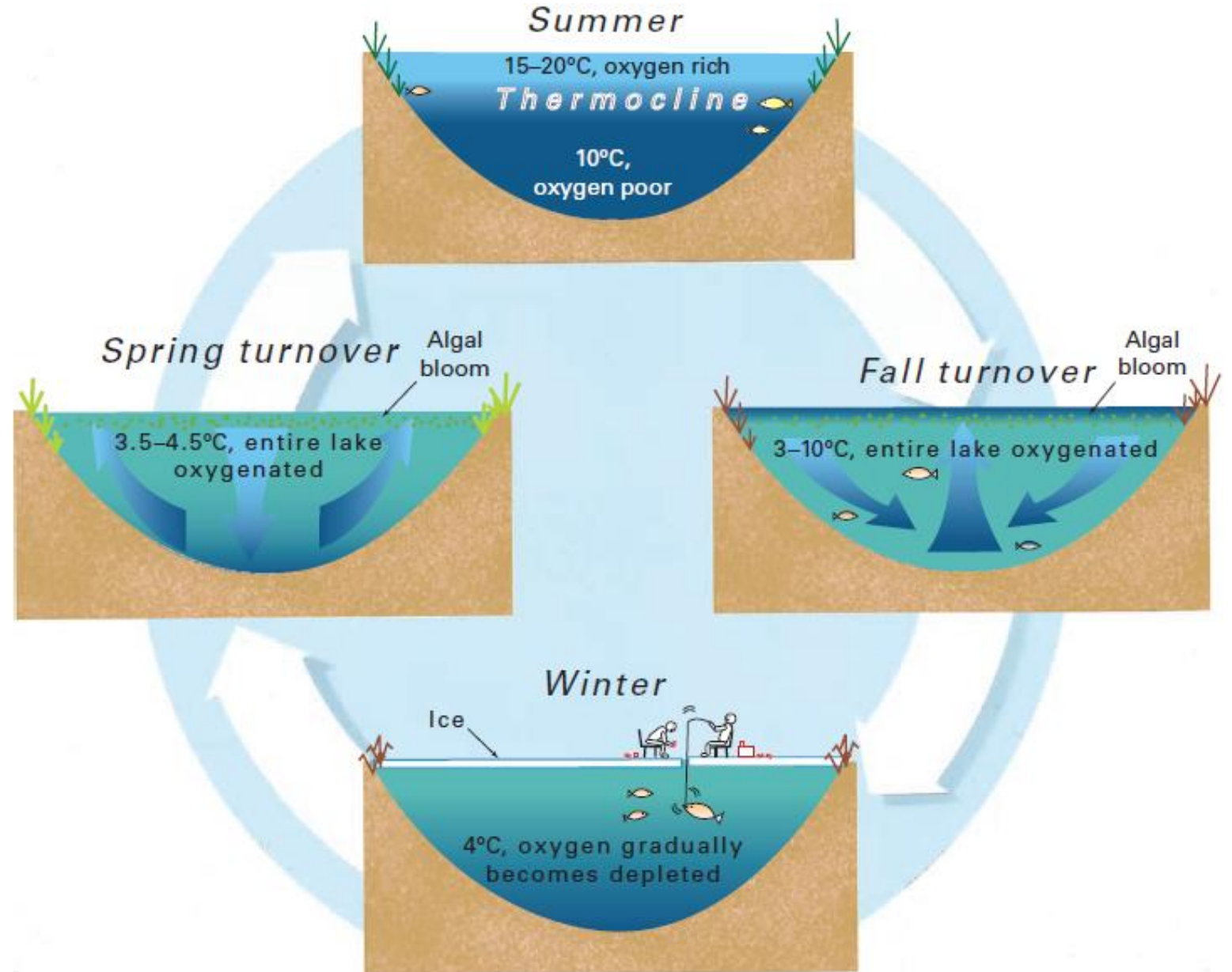


Temperature Layering and Turnover in Lakes



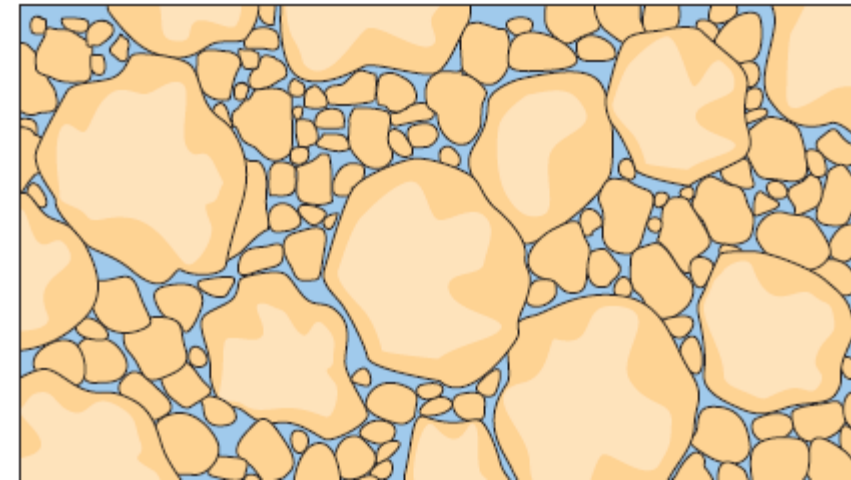
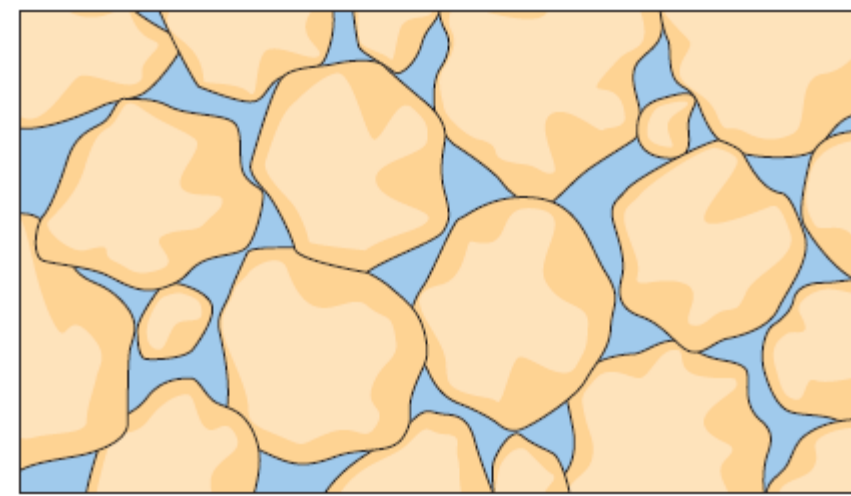
Temperature Layering and Turnover in Lakes

- Lakes in temperate climates develop temperature layering in both summer and winter.
- As a result, bottom waters become depleted in oxygen.
- In fall and spring, water temperature becomes constant throughout the lake and turnover brings new supplies of oxygen to the deep waters.



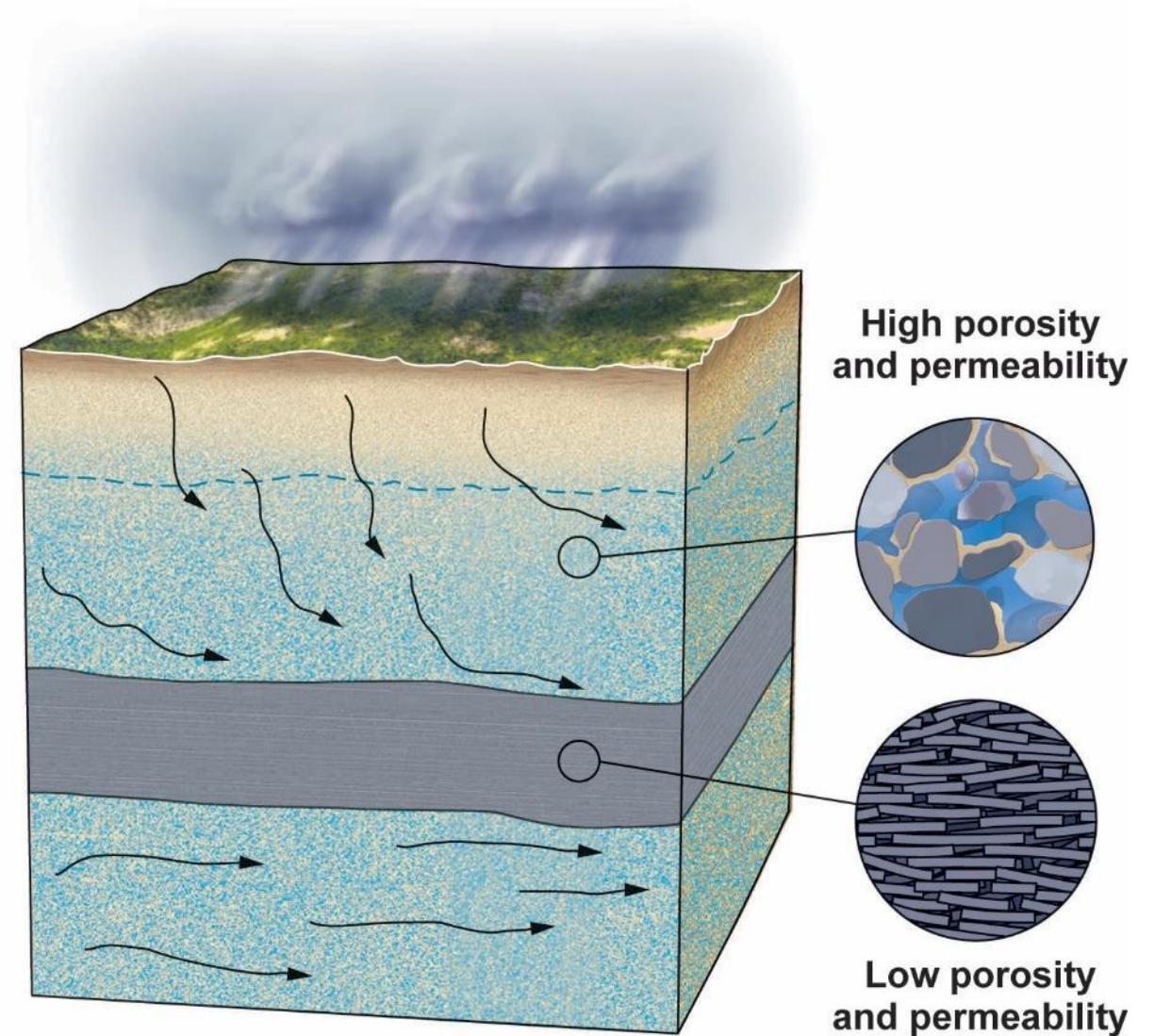
Groundwater- Porosity and Permeability

- Groundwater fills small cracks and voids in soil and bedrock.
- The volume percentage of these open spaces, called *pores*, is called **porosity**.
- Igneous and metamorphic rocks have very low porosities (a few percent or less) unless they are fractured. (Secondary porosity)
- Porosity indicates the amount of water that rock or soil can hold. In contrast, **permeability** refers to how interconnected the pore spaces are and relates to the ability of rock or soil to transmit water or any other fluid including petroleum and natural gas.



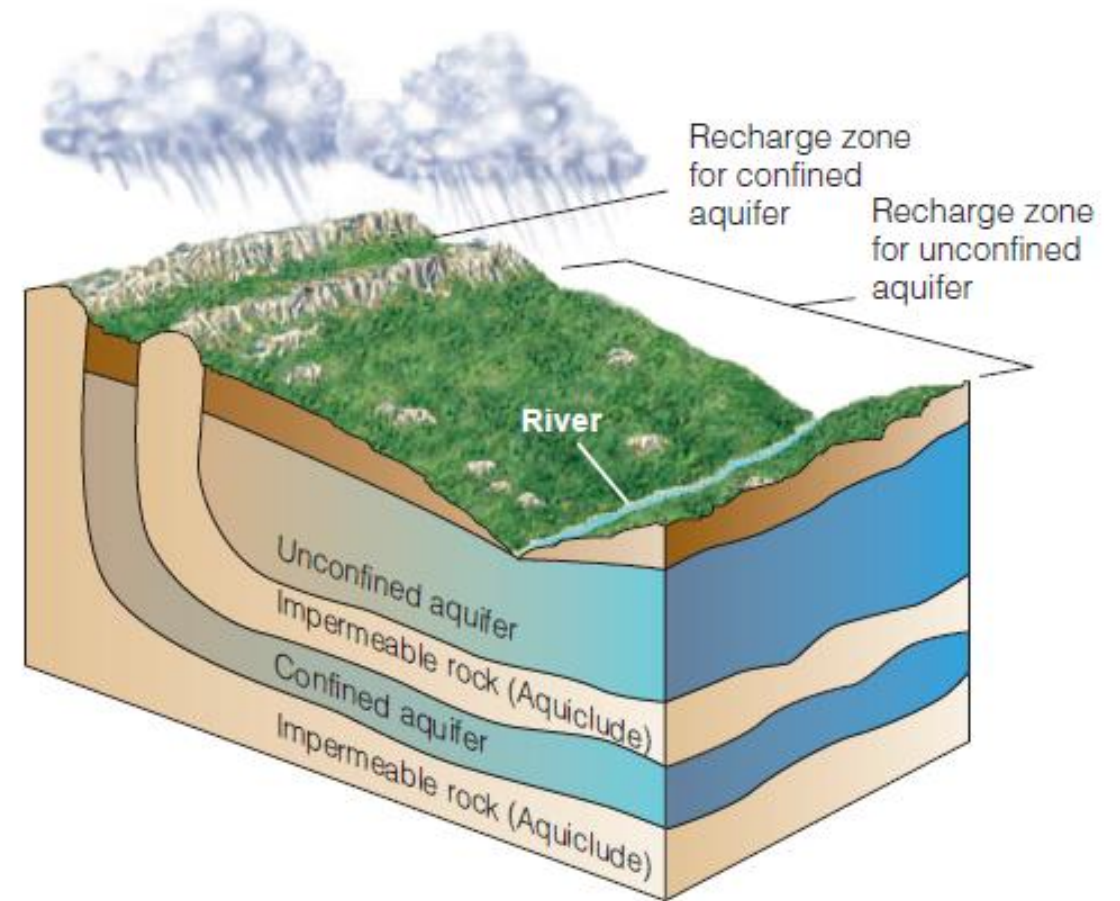
Aquifers and Aquitards

- Aquifers and aquitards are commonly interlayered.
- **A: aquifer**—sediment or rock that **transmits water easily**.
- both porous and permeable so that water flows into a well to replenish (recharge) water that is pumped out.
- Ex: Sand and gravel, sandstone, limestone, and highly fractured bedrock.
- **B: aquitard**—**impermeable or low permeability** sediment or rock that hinders water flow.
- Ex: Mudstone, clay, and unfractured igneous and metamorphic rocks.



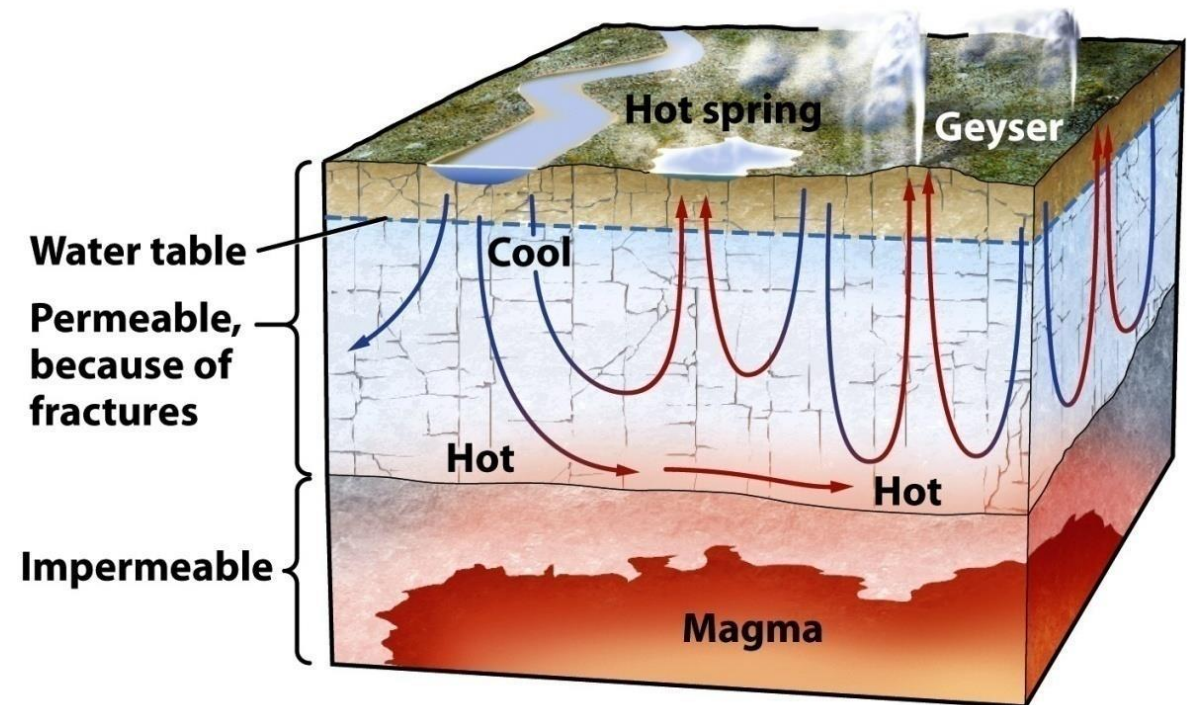
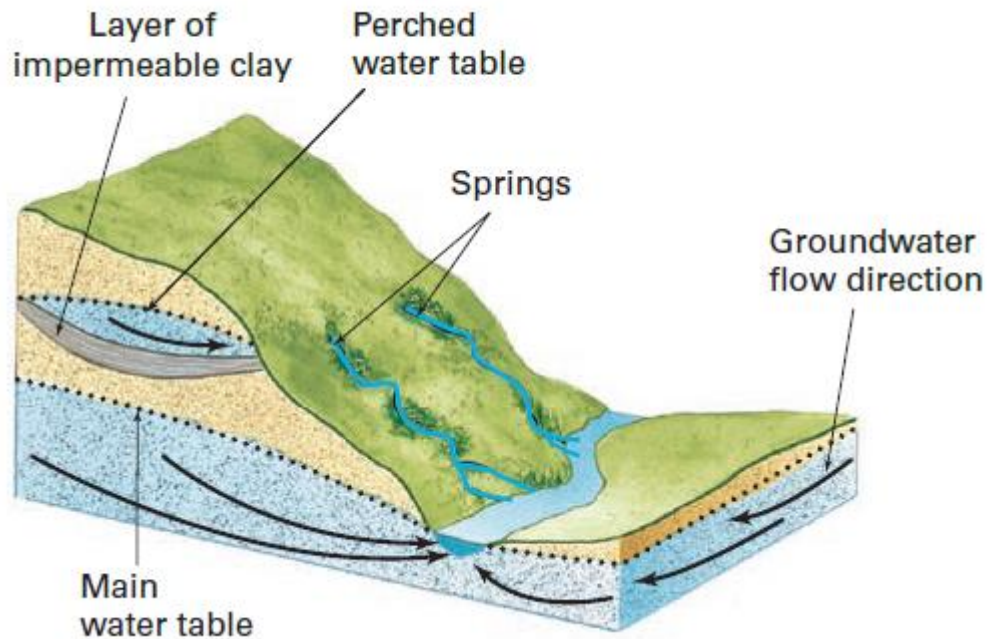
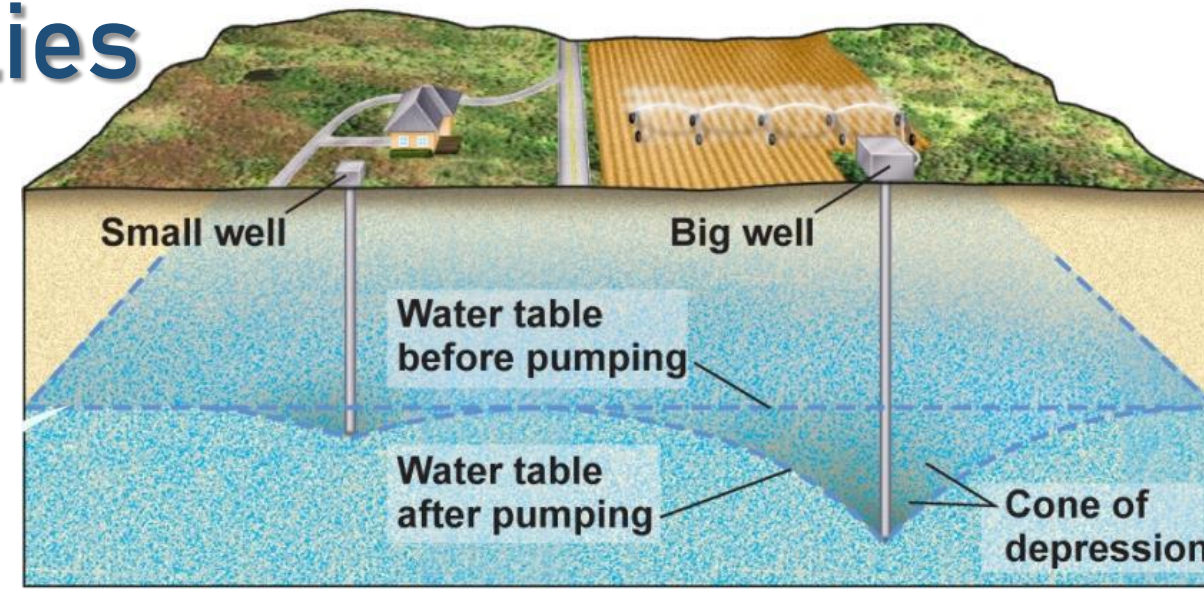
Aquifer

- An **aquifer** is an underground layer of rock, sand, gravel, or soil that **holds water** and allows it to flow
- **Unconfined**—an aquifer that intersects the surface and **open to the atmosphere** through pores in the rock and soil above the aquifer
 - In contact with the atmosphere
 - Easily contaminated
- **Confined**—water in a confined aquifer is trapped between impermeable rock layers
 - Isolated from the surface
 - Less susceptible to pollution

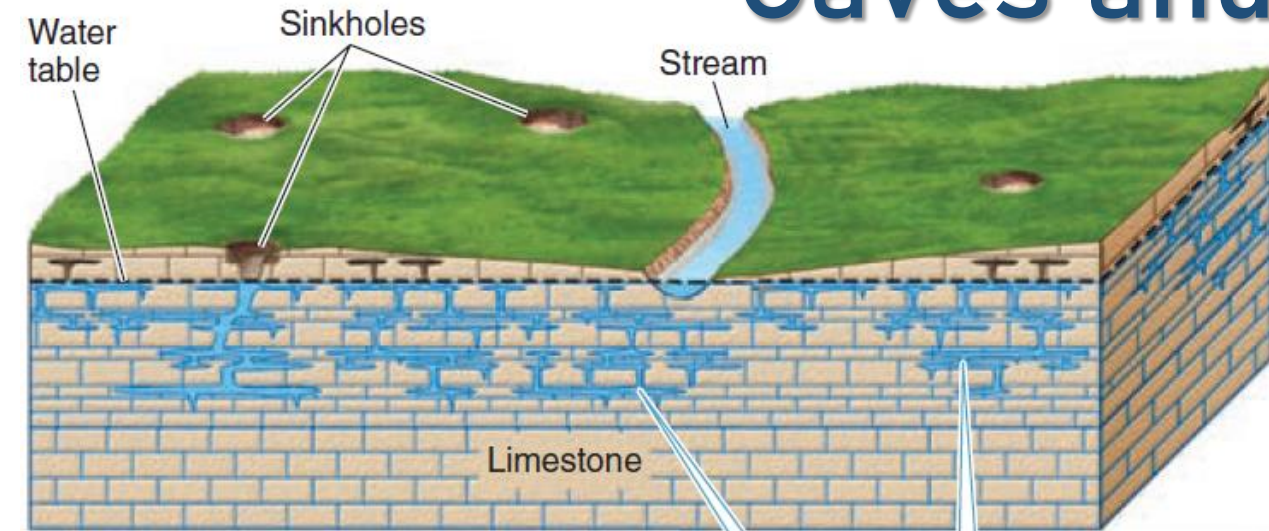


Tapping Groundwater Supplies

- Groundwater is obtained in two ways.
- Wells—holes excavated or drilled to obtain water.
- Springs—natural groundwater outlets
 - A **spring** occurs where the water table intersects the land surface and water flows or seeps onto the surface.



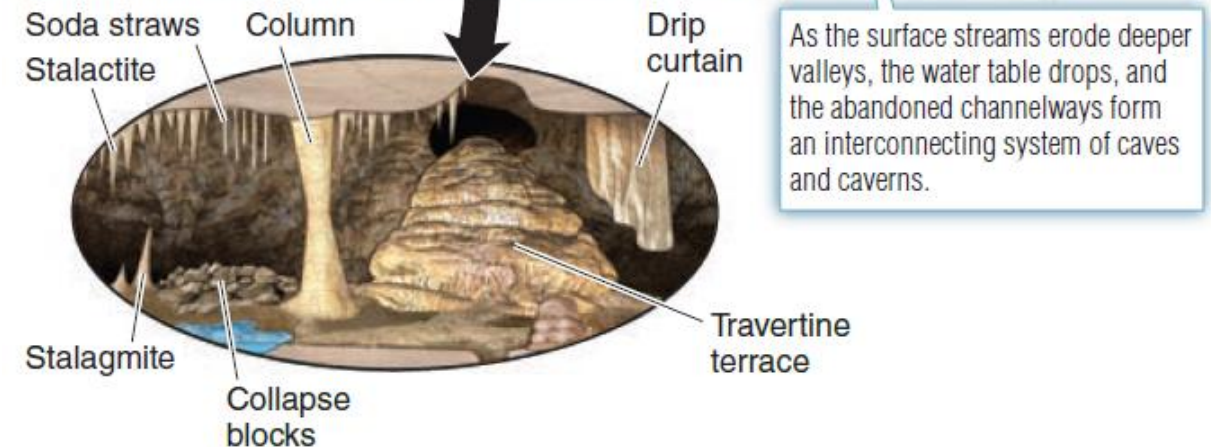
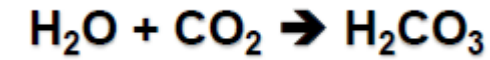
Caves and Sinkholes



As groundwater percolates through the zone of aeration and flows through the zone of saturation, it dissolves the carbonate rocks and gradually forms a system of passageways.



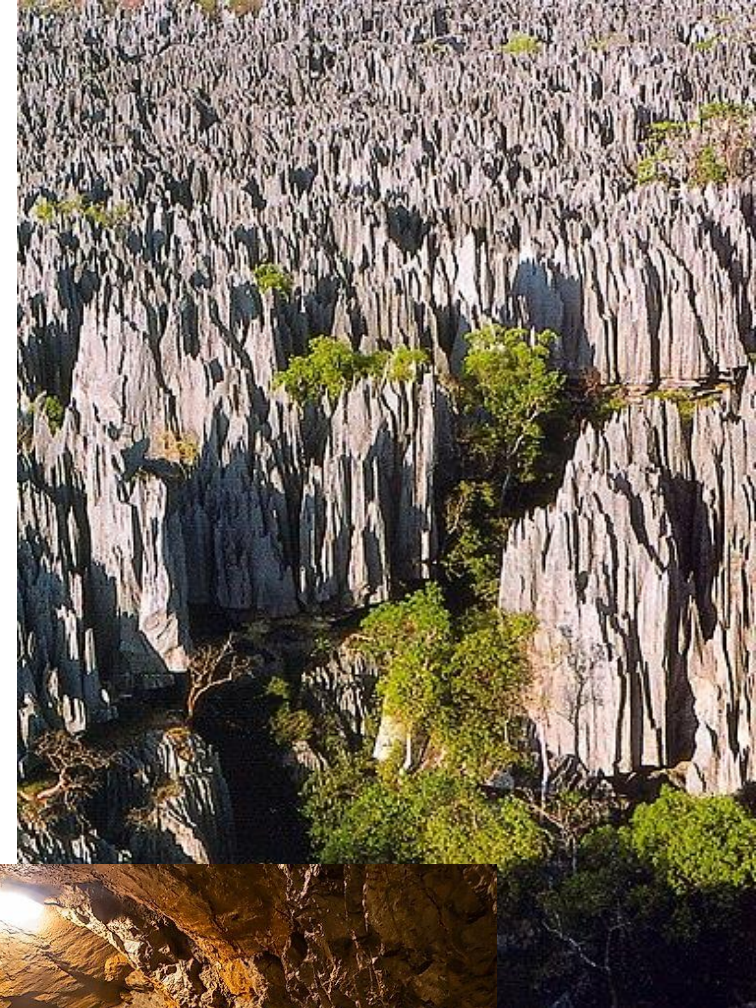
As the stream erodes more deeply, groundwater moves along the surface of the lower water table, forming a system of horizontal passageways through which dissolved rock is carried to the surface streams, thus enlarging the passageways.



As the surface streams erode deeper valleys, the water table drops, and the abandoned channelways form an interconnecting system of caves and caverns.

Karst topography

- In some regions of exceptionally **soluble rock**, sinkholes and caves are so numerous that they combine to form a distinctive topography characterized by many small, closed basins and intervening ridges or pinnacles.
- In this kind of landscape, streams disappear into the ground and eventually reappear elsewhere as large springs. Such terrain is called karst (German for bare, stony ground).
- It is often densely forested and has steep slopes.
- Karst landscapes cover some 20% of Earth's surface area and contain about 80% of its known caves.

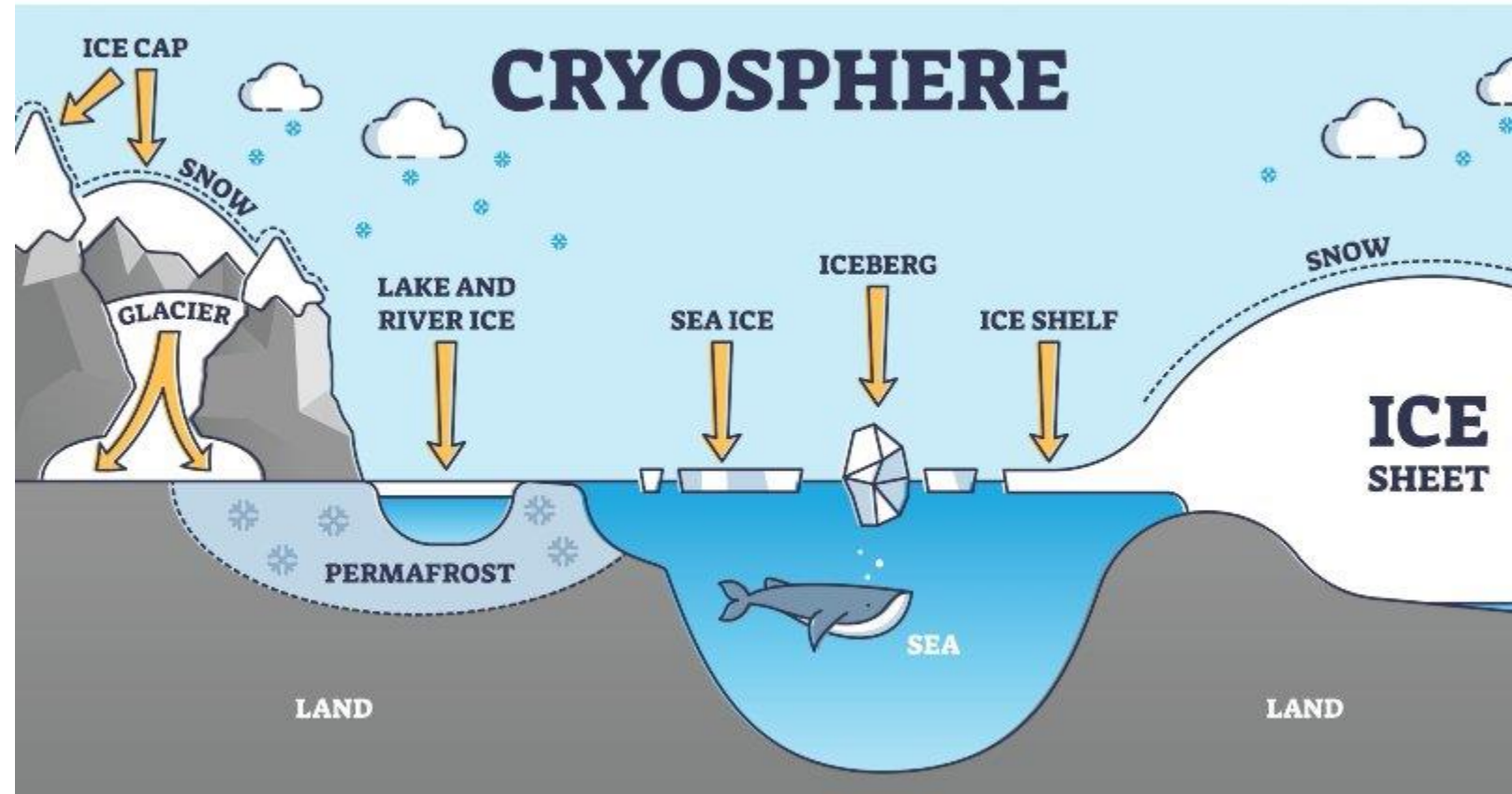


Cryosphere



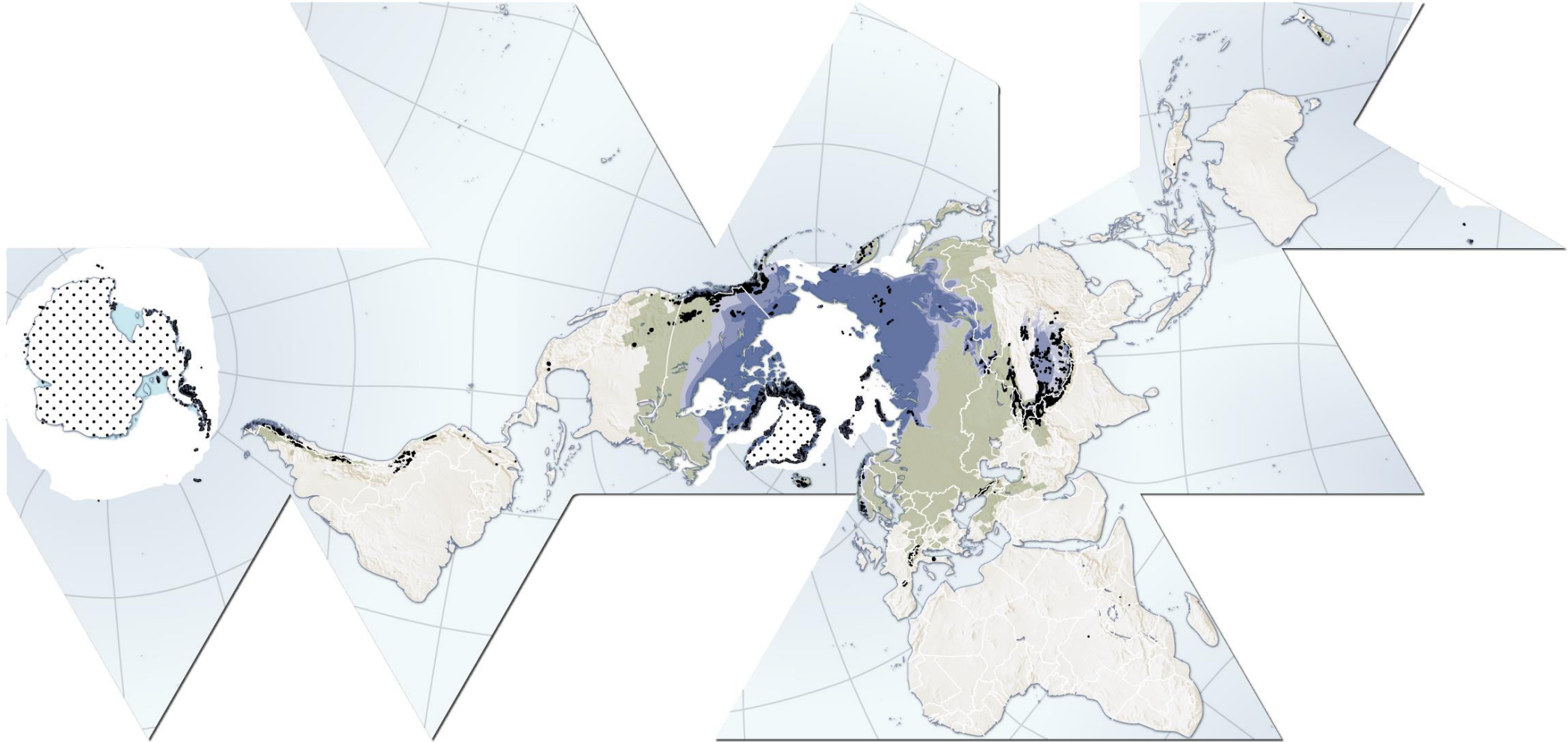
What is cryosphere ?

- The word cryosphere to describe the set of all locations on Earth where **water is found in solid form**, including areas of
 - Snow
 - Sea ice
 - Glaciers
 - Permafrost
 - Ice sheets and icebergs.
- The term “cryosphere” traces its origins to the Greek word 'kryos' for frost or ice cold.

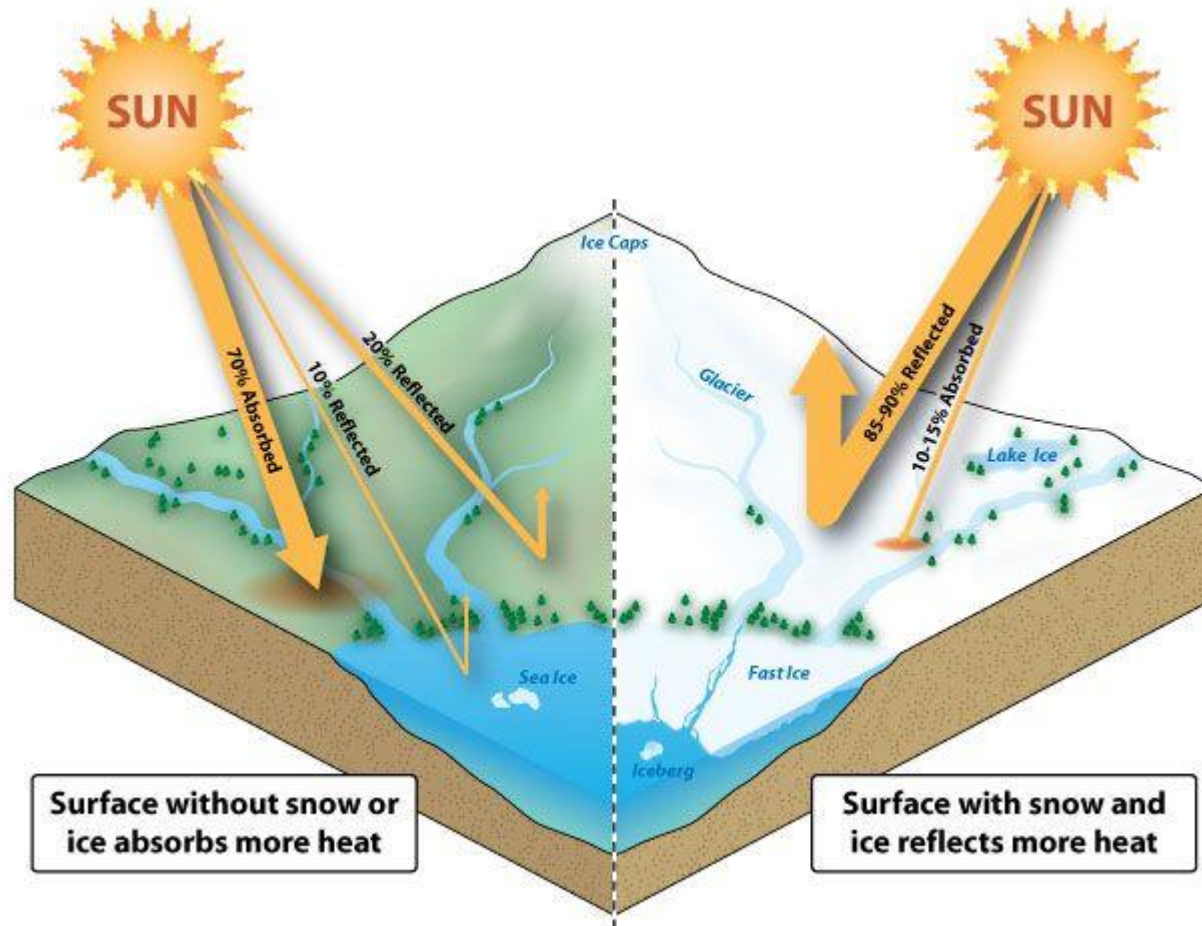


Where is cryosphere ?

- Snow
- Sea ice
- Ice shelves
- Ice sheets
- Glaciers and ice caps
- Permafrost, continuous
- Permafrost, discontinuous
- Permafrost, isolated



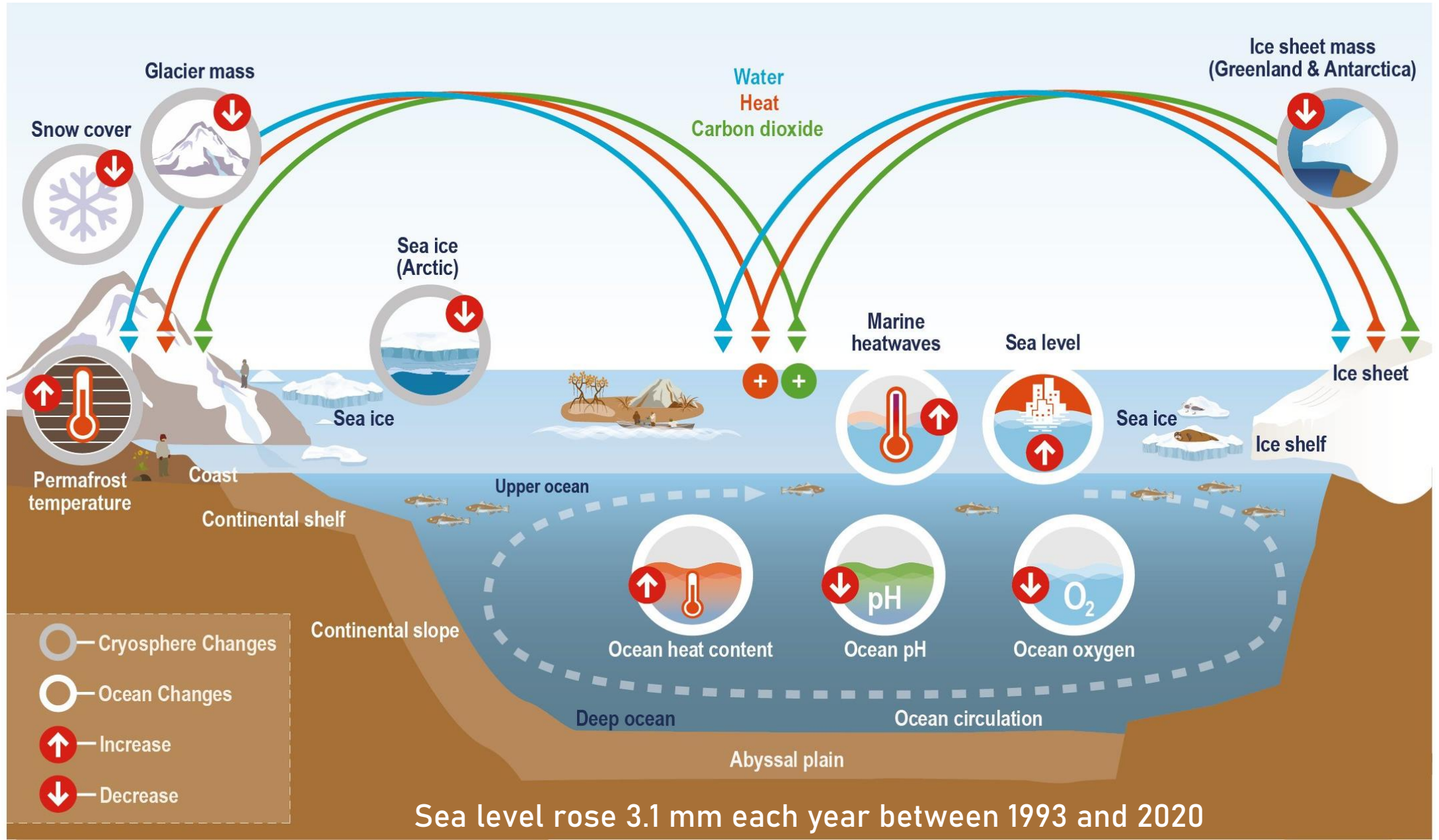
Why do we care ?



- Change in climate system
e.g.: Increase in temperature, release of methane/ CO_2
- Change in ocean and atmospheric circulation.
- Changes in ecosystem, that depend on ice and snow.

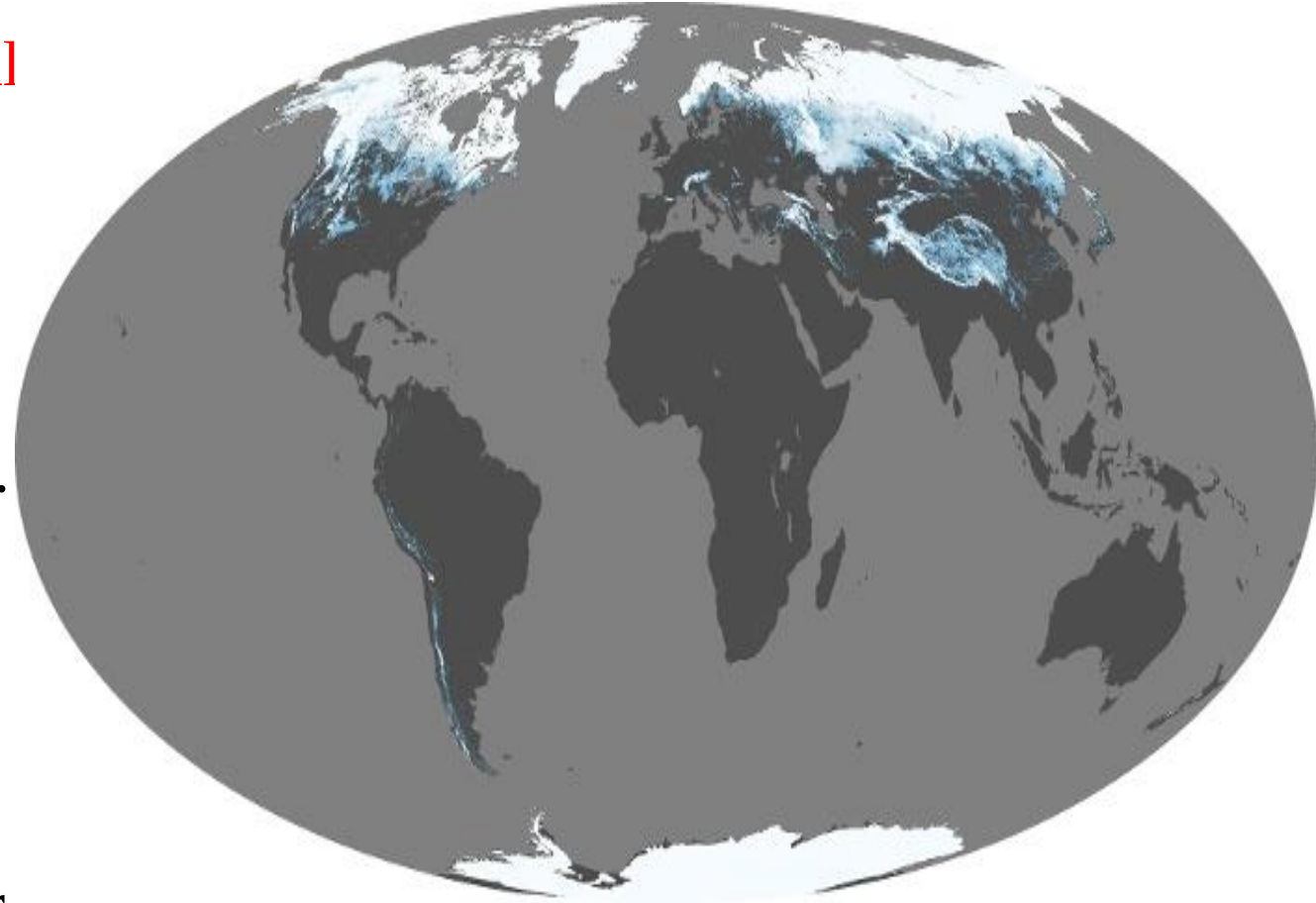


Why do we care ?



Snow

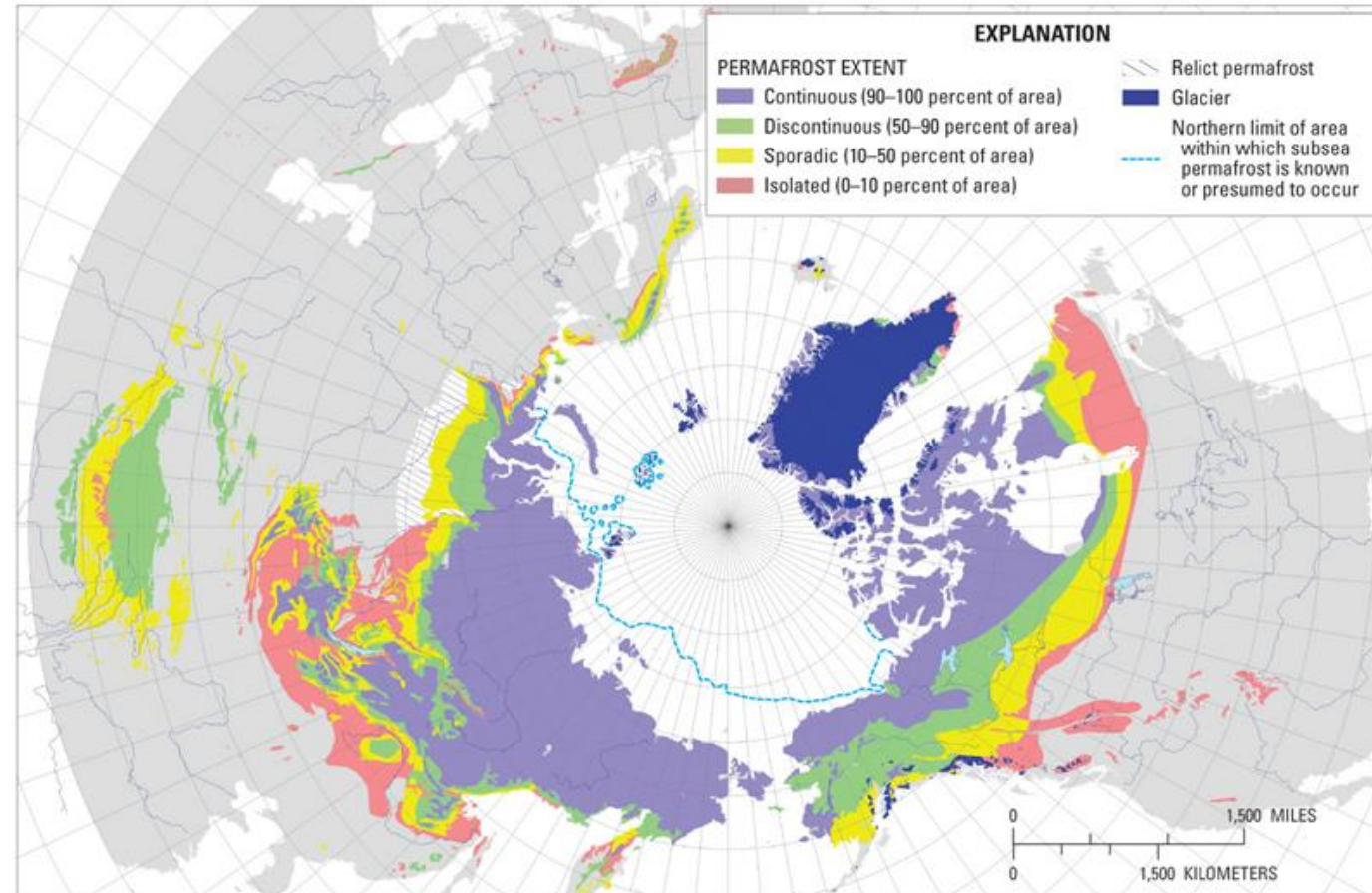
- Seasonal snow cover has the **largest spatial extent** (not volume) of components of cryosphere.
- In the northern hemisphere, almost one-fourth (25%) of land area is covered by snow and frozen ground during the winter.
- The highly reflective (80 – 90%) surface of snow bounces sunlight back into space, thereby reducing surface air temperature.
- When snow melts, it becomes a major source of water for rivers and moisture for agricultural soils.



Permafrost

- **Permafrost** (i.e., **permanently frozen ground**), which is soil, sediment, or rock that remains at or below 0°C for at least two years.
- Can be up to 1000 m deep.
- **Permafrost is mostly confined to the Northern Hemisphere**, where it covers approximately 11% of the hemisphere's land surface.
- That means 18 million km^2 in the Northern Hemisphere is permafrost.
- Most of it is originated during glacial times (> 20000 years).
- Contains a lot of organic material which can decay as permafrost thaws releasing CO_2 and methane.

Permafrost Extent in the Northern Hemisphere



Glaciers



- A **glacier** is a large, slow-moving mass of ice that forms on land from compacted snow over many years
- They **flow downhill** under their own weight, like a very slow river of ice
- Form in places where more snow falls in winter than it melts in summer (mountains, poles)

<https://www.youtube.com/watch?v=ghC-Ut0fW4o> - Glacier movement



Sea ice

- Approximately **two thirds of the area** of Earth's persistent **ice cover floats as a thin sea ice** on polar oceans.
- Sea ice drifts many km/day. Average drift rates in the Arctic Ocean are about 7 km/day, whereas in the Greenland and Bering seas velocities reach 15 km/day.
- **An interesting feature of sea ice is that it consists of fresh water; the salt that is characteristic of seawater is excluded from the ice crystals as they form.**
- **Does melting sea ice rise sea level?**
- **Since sea ice floats on water, the melting of sea ice has no effect on sea level.**



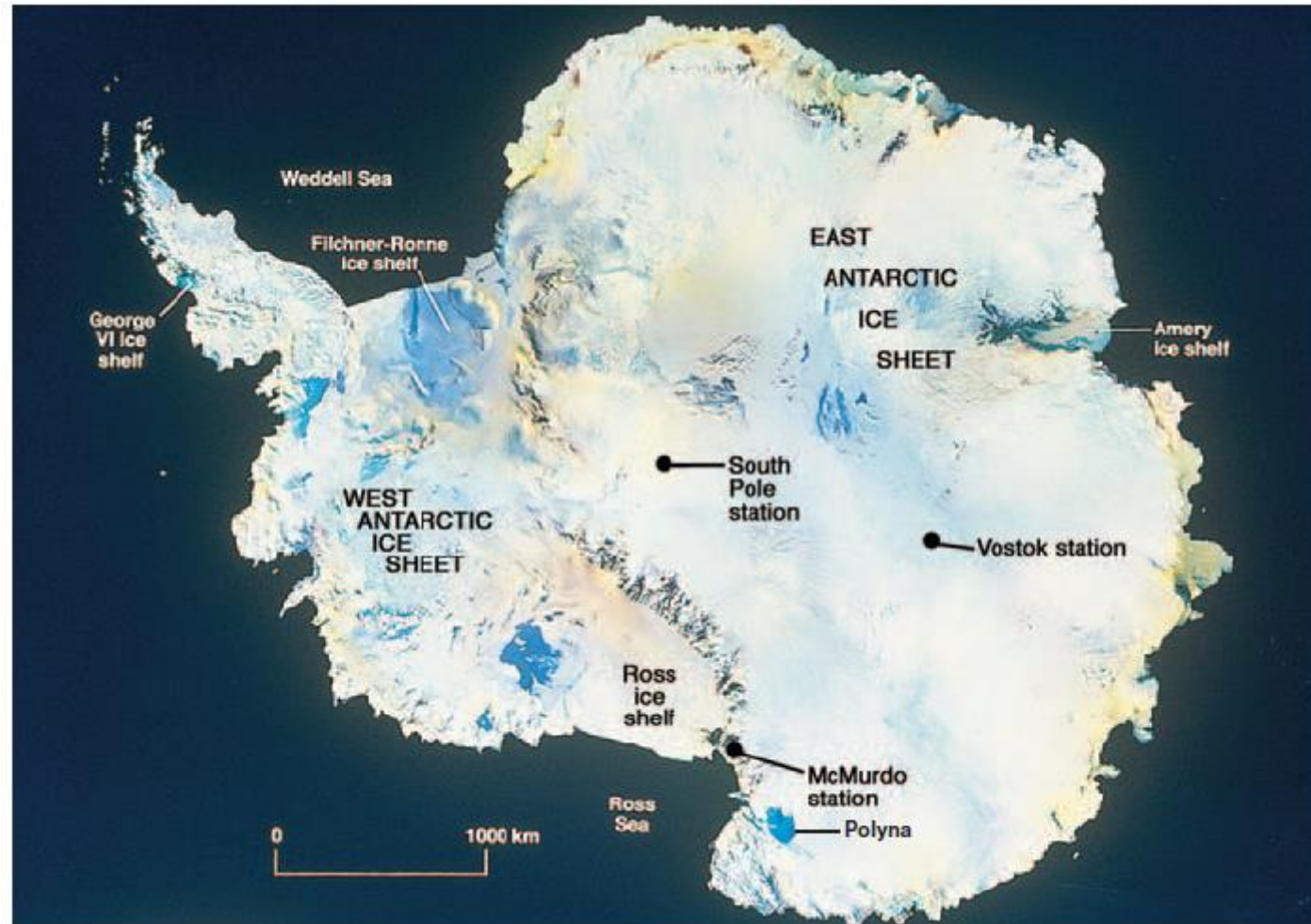
<https://www.youtube.com/watch?v=JJHaYGsUe78> – Icy Fingers of Death

Ice sheet

- Ice sheet, or **continental glacier** A glacier that covers an area of 50,000 square kilometers or more and spreads outward in all directions under its own weight.
- These two ice sheets contain 99 percent of the world's ice and about three-fourths of Earth's freshwater.
- The Greenland sheet is more than 3.3 kilometers thick in places and covers 1.7 million square kilometers.
- Yet it is small compared with the Antarctic ice sheet, which blankets nearly 14 million square kilometers, roughly the size of the contiguous United States and Mexico combined.
- The Antarctic ice sheet covers entire mountain ranges, and the mountains that rise above its surface are islands of rock in a sea of ice.



Ice shelves



Ice Sheets = on land

Ice Shelves = floating on ocean, but still attached to land ice